# Lecture 3

## Integration by Parts

Our primary goal is how to integrate product of two functions.

For example,

$$\int f(x)\, g(x)\, dx$$

Here we use a formula to integrate product of two functions which we call it integration by parts.

Say $u = f(x)$ $\quad du = f'(x)\,dx$

$g(x)\,dx = dv$

$$\boxed{\int u\,dv = uv - \int v\,du}$$

$$\int u\,dv + \int v\,du = uv$$

**Example:** Evaluate: $\int x \cos x\, dx$

Say, $u = x$ and $dv = \cos x\, dx$

$du = dx$ $\quad v = \sin x$

$$\int x \cos x\, dx = x \sin x - \int \sin x\, dx$$

$$= x \sin x - (-\cos x) + C$$

$$= x \sin x + \cos x + C$$



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**Example:** Evaluate $\int e^x \cos x \, dx$

L – Logarithmic
I – Inverse trigonometric
A – Algebric
T – Trigonometric
E – Exponential

**Sol^n.**

Say, $u = \cos x$ | $dv = e^x dx$

$du = -\sin x \, dx$ | $v = e^x$

Thus,

$$\int e^x \cos x \, dx = e^x \cos x - \int e^x (-\sin x) \, dx$$

$$\therefore I = e^x \cos x + \int e^x \sin x \, dx \qquad \text{——— (1)}$$

Ø Say, $I = \int e^x \cos x \, dx$

Now let $\sin x = u$ $\qquad dv = e^x dx$

$\cos x \, dx = du$ $\qquad v = e^x$

$$\int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx = e^x \sin x - I$$

From (1) $\quad I = e^x \cos x + e^x \sin x - I$

$$2I = e^x \cos x + e^x \sin x$$

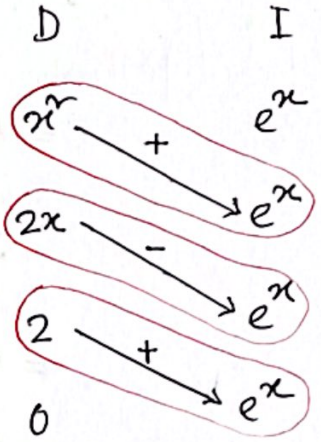
$$I = \frac{e^x}{2}(\cos x + \sin x)$$

Therefore $\int e^x \cos x \, dx = \frac{1}{2} e^x (\sin x + \cos x) + c$ ∴

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We can also use shortcut techniques for ~~pro~~ integration by parts.

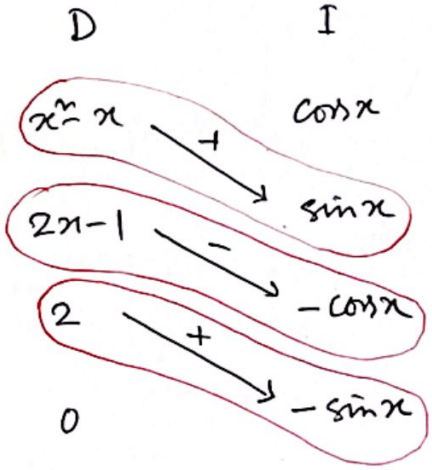
Example: Evaluate: $\int x^2 \cdot e^x \, dx$




Sol$^n$

$$\int x^2 e^x dx = x^2 e^x - 2xe^x + 2e^x + c$$

Example: Evaluate: $\int (x^2 - x)\cos x \, dx$




Sol$^n$

$$\int (x^2 - x)\cos x \, dx$$

$$= (x^2 - x)\sin x + (2x-1)(-)(-\cos x) + 2(+)\cdot(-\sin x)$$

$$= (x^2 - x)\sin x + (2x-1)\cos x - 2\sin x + C$$

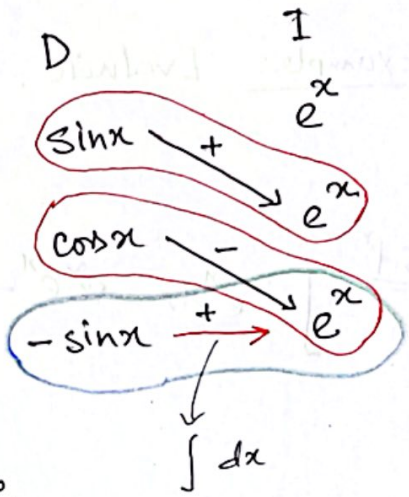
Multiply ~~under~~ inside red color's function with sign and use '+' inbetween two ~~red~~ bounded red color's mark.

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**Example:** Evaluate $\int e^x \sin x \, dx$

Sol:

Say, $I = \int e^x \sin x \, dx$




$= e^x \sin x - e^x \cos x - \int e^x \sin x \, dx$

$= e^x \sin x - e^x \cos x - I$

$\Rightarrow 2I = e^x \sin x - e^x \cos x + C_1$

$I = \frac{1}{2} e^x [\sin x - \cos x] + C$ ※

## TRY YOURSELF

Evaluate:

1. $\int x \ln x \, dx$
2. $\int e^{3x} \cos 2x \, dx$
3. $\int x \tan^2 x \, dx$
4. $\int (x + x\cos x) \, dx$
5. $\int x^3 e^{x^2} \, dx$

## EXTRA PROBLEMS:

Exercise : 7.2

Page: 498

PROBLEMS: 1-38 (TRY ODD NUMBERS)

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# Lecture 4

## Integrating Powers of Sine and Cosine

1. $\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x\,dx$

2. $\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1}x\sin x + \frac{n-1}{n}\int \cos^{n-2}x\,dx$

We can prove the above integration by reduction formulas.

$uv = \int u\,dv + \int v\,du$

$\int u\,dv = uv - \int v\,du$

**For (2)**

$$\int \cos^n x\,dx = \int \cos^{n-1}x\cos x\,dx$$

Say, $u = \cos^{n-1}x$ $\qquad dv = \cos x\,dx$

$du = (n-1)\cos^{n-2}x(-\sin x)dx$ $\qquad v = \sin x$

So that

$$\int \cos^n x\,dx = \cos^{n-1}x\sin x + (n-1)\int \sin^2 x\cos^{n-2}x\,dx$$

$$= \cos^{n-1}x\sin x + (n-1)\int (1-\cos^2 x)\cos^{n-2}x\,dx$$

$$= \cos^{n-1}x\sin x + (n-1)\int \cos^{n-2}x\,dx - (n-1)\int \cos^n x\,dx$$

$$(n-1+1)\int \cos^n x\,dx = \cos^{n-1}x\sin x + (n-1)\int \cos^{n-2}x\,dx$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1}x\sin x + \frac{n-1}{n}\int \cos^{n-2}x\,dx$$

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Example: $\int \sin^3 x\,dx$ , $n=3$

$$= -\frac{1}{3}\sin^{2}x\cos x + \frac{3-1}{3}\int \sin x\,dx$$

$$= -\frac{1}{3}(1-\cos^2 x)\cos x + \frac{2}{3}(-\cos x) + C$$

$$= -\frac{1}{3}\cos x + \frac{1}{3}\cos^3 x - \frac{2}{3}\cos x + C$$

$$= \frac{1}{3}\cos^3 x - \cos x + C$$

$\therefore$

## Integrating Products of Sines and Cosines

$$\int \sin^m x \cos^n x\,dx$$

If n odd then $\cos^2 x = 1-\sin^2 x$ ; say $\Rightarrow$ $u = \sin x$

If m odd then $\sin^2 x = 1-\cos^2 x$ ; say, $u = \cos x$

If m even, n even then $\sin^2 x = \frac{1}{2}(1-\cos 2x)$

$\cos^2 x = \frac{1}{2}(1+\cos 2x)$

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**Example:** Evaluate $\int \sin^4 x \cos^5 x \, dx$

**Sol$^n$:**

$$\int \sin^4 x \cos^5 x \, dx = \int \sin^4 x (1-\sin^2 x)^2 \cos x \, dx$$

$$= \int u^4 (1-u^2)^2 \, du$$

$$= \int u^4 (1 - 2u^2 + u^8) \, du$$

$$= \int (u^4 - 2u^6 + u^8) \, du$$

$$= \frac{u^5}{5} - 2\frac{u^7}{7} + \frac{u^9}{9} + C$$

$$= \frac{1}{5}\sin^5 x - \frac{2}{7}\sin^7 x + \frac{1}{9}\sin^9 x + C \quad \square$$

~~$u = \cos x$~~
~~$du = -\sin x$~~

Let $u = \sin x$
$du = \cos x \, dx$

**Extra Problem:**

1. $\int \sin^2 t \cos^3 t \, dt$
2. $\int \sin^3 x \cos^3 x \, dx$
3. $\int \sin^4 3x \cos^3 3x \, dx$
4.

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## Trigonometric Identities

$$2\sin\alpha\cos\beta = \sin(\alpha-\beta) + \sin(\alpha+\beta)$$

$$2\sin\alpha\sin\beta = \cos(\alpha-\beta) - \cos(\alpha+\beta)$$

$$2\cos\alpha\cos\beta = \cos(\alpha-\beta) + \cos(\alpha+\beta)$$

**Example:** Evaluate $\int \sin 7x \cos 3x \, dx$

$$= \frac{1}{2}\int [\sin(7x-3x) + \sin(7x+3x)]\, dx$$

$$= \frac{1}{2}\int [\sin 4x + \sin 10x]\, dx$$

$$= \frac{1}{2}\left[ -\frac{\cos 4x}{4} - \frac{\cos 10x}{10} \right] + C$$

$$= -\frac{1}{8}\cos 4x - \frac{1}{20}\cos 10x + C$$

## EXTRA PROBLEMS:

Book – H. Anton

page: 506 - 507

PROBLEMS: 1 – 22

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* $\boxed{\int \tan^m x \sec^n x \, dx}$

If $n$ even then $\sec^2 x = \tan^2 x + 1$ say, $u = \tan x$

If $m$ odd then $\tan^2 x = \sec^2 x - 1$ say, $u = \sec x$

If $\begin{cases} m \text{ even} \\ n \text{ odd} \end{cases}$ then $\tan^2 x = \sec^2 x - 1$

Example: Evaluate $\int \tan^2 x \sec^4 x \, dx$

$$= \int \tan^2 x \sec^2 x \cdot \sec^2 x \, dx$$

$$= \int \tan^2 x (\tan^2 x + 1) \sec^2 x \, dx$$

$$= \int u^2 (u^2 + 1) \, du$$

$$= \frac{u^5}{5} + \frac{u^3}{3} + C$$

$$= \frac{1}{5} \tan^5 x + \frac{1}{3} \tan^3 x + C$$

Say,
$u = \tan x$
$du = \sec^2 x \, dx$

□

Extra:

① $\int \tan^5 x \sec^4 x \, dx$

② $\int \tan^2 x \sec^3 x \, dx$

③ $\int \sec^5 x \, dx$

④ $\int \tan^4 x \, dx = \int \tan^2 x (\sec^2 x - 1) \, dx$

⑤ BOOK – H. Anton
page: 507
PROBLEMS: 23 – 50

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**Problem:** Evaluate $\int \sec^4 x\, dx$

**Sol$^n$:**

$$\int \sec^4 x\, dx = \int \sec^2 x \cdot \sec^2 x\, dx$$

$$= \int (1+\tan^2 x)\sec^2 x\, dx \qquad \tan x = u, \quad \sec^2 x\, dx = du$$

$$= \int (1+u^2)\, du$$

$$= u + \frac{1}{3}u^3 + C$$

$$= \tan x + \frac{1}{3}\tan^3 x + C$$

✳

**Problem:** Evaluate: $\int \tan^4 x\, dx$.

**Sol$^n$:**

$$\int \tan^4 x\, dx = \int \tan^2 x \cdot \tan^2 x\, dx = \int \tan^2 x\,(\sec^2 x - 1)\, dx$$

$$= \int \tan^2 x \sec^2 x\, dx - \int \tan^2 x\, dx$$

$$= \int \tan^2 x \sec^2 x\, dx - \int (\sec^2 x - 1)\, dx$$

$$= \int u^2\, du - \tan x + x + C_1 \qquad \tan x = u, \quad \sec^2 x\, dx = du$$

$$= \frac{1}{3}u^3 - \tan x + x + C$$

$$= \frac{1}{3}\tan^3 x - \tan x + x + C$$

✳

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**Problem:** $\int \sec^3 x\,dx$

$= \int \sec x \sec^2 x\,dx$

| D | | I |
|---|---|---|
| $\sec x$ | $+$ | $\sec^2 x$ |
| $\sec x \tan x$ | $-$ | $\tan x$ |

So, $\int \sec^3 x\,dx = \sec x \tan x - \int \sec x \tan^2 x\,dx$

$= \sec x \tan x - \int \sec x(\sec^2 x - 1)\,dx$

$= \sec x \tan x - \int \sec^3 x\,dx + \int \sec x\,dx$

$\therefore 2\int \sec^3 x\,dx = \sec x \tan x + \int \sec x\,dx$

$= \sec x \tan x + \ln|\sec x + \tan x|$

Hence, $\int \sec^3 x\,dx = \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$

**Problem:** Evaluate $\int \sec^5 x\,dx$

Sol: $\int \sec^5 x\,dx = \int \sec^3 x\, d\sec^2 x\,dx$

| D | | I |
|---|---|---|
| $\sec^3 x$ | $+$ | $\sec^2 x$ |
| $3\sec^2 x \sec x \tan x$ | $-$ | $\tan x$ |

$= \sec^3 x \tan x - 3\int \sec^3 x \tan^2 x\,dx$

$= \sec^3 x \tan x - 3\int \sec^3 x(\sec^2 x - 1)\,dx$

$= \sec^3 x \tan x - 3\int \sec^5 x\,dx + 3\int \sec^3 x\,dx$

$\therefore 4\int \sec^5 x\,dx = \sec^3 x \tan x + 3\int \sec^3 x\,dx$

$\int \sec^5 x\,dx = \frac{1}{4}\sec^3 x \tan x + \frac{3}{4}\left[\frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x|\right]$

$= \frac{1}{4}\sec^3 x \tan x + \frac{3}{8}\sec x \tan x + \frac{3}{8}\ln|\sec x + \tan x| + C$

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## Trigonometric Substitutions:

| | | Restriction |
|---|---|---|
| 1. If $\sqrt{a^2-x^2}$, | say $x = a\sin\theta$ | $-\pi/2 \le \theta \le \pi/2$ |
| 2. If $\sqrt{a^2+x^2}$, | say $x = a\tan\theta$ | $-\pi/2 < \theta < \pi/2$ |
| 3. If $\sqrt{x^2-a^2}$, | say $x = a\sec\theta$ | $\begin{cases} 0 \le \theta < \pi/2 & \text{if } x \ge a \\ \pi/2 < \theta \le \pi & \text{if } x \le -a \end{cases}$ |

**Problem:** Evaluate $\int \frac{dx}{\sqrt{4+x^2}}$

$x = 2\tan\theta \quad ; -\pi/2 < \theta < \pi/2$

$dx = 2\sec^2\theta\, d\theta$

$4+x^2 = 4+4\tan^2\theta = 4(1+\tan^2\theta)$

$= 4\sec^2\theta$

$\sqrt{\sec^2\theta} = |\sec\theta|$

$\therefore \int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2\sec^2\theta\, d\theta}{\sqrt{4\sec^2\theta}}$

$= \int \frac{2\sec^2\theta\, d\theta}{2|\sec\theta|}$

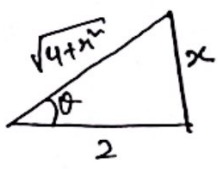
$= \int \sec\theta\, d\theta$

$= \ln|\sec\theta + \tan\theta| + C$

$= \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C$

$\tan\theta = \frac{x}{2} = \frac{\text{opp}}{\text{adj}}$

$\sec\theta = \frac{\sqrt{4+x^2}}{2} = \frac{\text{hyp}}{\text{adj}}$




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**Example:** Evaluate $\int \frac{dx}{x^2\sqrt{4-x^2}}$

Soln:

$$\int \frac{dx}{x^2\sqrt{4\left(1-\frac{x^2}{4}\right)}}$$

$$= \frac{1}{2}\int \frac{dx}{x^2\sqrt{1-\left(\frac{x}{2}\right)^2}}$$

$$= \frac{1}{2}\int \frac{2\cos\theta\, d\theta}{4\sin^2\theta\sqrt{1-\sin^2\theta}}$$

$$= \frac{1}{4}\int \frac{\cos\theta\, d\theta}{\sin^2\theta\sqrt{\cos^2\theta}}$$

$$= \frac{1}{4}\int \frac{1}{\sin^2\theta}\, d\theta$$

$$= \frac{1}{4}\int \csc^2\theta\, d\theta$$

$$= \frac{1}{4}(-\cot\theta) + C$$

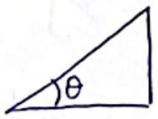
$$= -\frac{1}{4}\cot\theta + C$$

$$= -\frac{1}{4}\frac{\sqrt{4-x^2}}{x} + C$$

wt

$\frac{x}{2} = \sin\theta$, $x = 2\sin\theta$

$dx = 2\cos\theta\, d\theta$




$\sin\theta = \frac{x}{2} = \frac{opp}{hyp.}$

$adj = \sqrt{4-x^2}$

$\cot\theta = \frac{adj}{opp} = \frac{\sqrt{4-x^2}}{x}$

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**Example:** $\int \frac{\sqrt{x^2-9}}{x}\,dx$

$\frac{x}{3} = \sec\theta \qquad x = 3\sec\theta$

$\frac{1}{3}dx = \sec\theta\tan\theta\,d\theta$

$= \int \frac{3\sqrt{(\frac{x}{3})^2-1}}{x}\,dx$

$= \cancel{3}\int \frac{\sqrt{\sec^2\theta-1}}{\cancel{x}/\cancel{3}}\,dx$

$= 3\int \frac{\sqrt{\sec^2\theta-1}}{\cancel{3}\sec\theta}\cdot \cancel{3}\sec\theta\tan\theta\,d\theta$

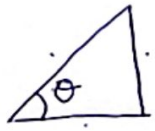
$= 3\int \tan\theta\cdot\tan\theta\,d\theta$

$= 3\int \tan^2\theta\,d\theta$

$= 3\int (\sec^2\theta-1)\,d\theta$

$= 3\left[\tan\theta-\theta\right]+c$

$= 3\left[\frac{\sqrt{x^2-9}}{3} - \tan^{-1}\left(\frac{\sqrt{x^2-9}}{3}\right)\right]+c$




$\sec\theta = \frac{x}{3} = \frac{\text{hyp.}}{\text{adj}}$

$\text{opp} = \sqrt{x^2-9}$

$\tan\theta = \frac{\text{opp.}}{\text{adj}} = \frac{\sqrt{x^2-9}}{3}$

$\theta = \tan^{-1}\frac{\sqrt{x^2-9}}{3}$

or

$\sec^{-1}\left(\frac{x}{3}\right)$

**Problem :** Evaluate : $\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}}$

**Soln** We already solve it without limit and the finally we get

$$\int \frac{dx}{x^2\sqrt{4-x^2}} = -\frac{1}{4}\cot\theta + c$$

$\sin = a$

~~$\sin\theta = \frac{x}{2}$~~

$x = 2\sin\theta$

$\theta = \sin^{-1}(x/2)$

If we use $\theta$ as limit then we must change the limit of $\theta$ instead of limit of $x$. So

| $x$ | 1 | $\sqrt{2}$ |
|---|---|---|
| $\theta$ | $\sin^{-1}(1/2)$ = $\pi/6$ | $\sin^{-1}(1/\sqrt{2})$ = $\pi/4$ |

Thus.

$$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}} = -\frac{1}{4}(\cot\theta)\Big|_{\pi/6}^{\pi/4}$$

$$= -\frac{1}{4}(\cot\pi/4 - \cot\pi/6)$$

$$= -\frac{1}{4}(1-\sqrt{3}) = \frac{1}{4}(\sqrt{3}-1) \quad \text{※}$$

OR

Without changing the limit, we also use the alternate answer. i.e.,

$$\int \frac{dx}{x^2\sqrt{4-x^2}} = -\frac{1}{4}\frac{\sqrt{4-x^2}}{x} + c$$

So, $$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}} = -\frac{1}{4}\left(\frac{\sqrt{4-x^2}}{x}\right)\Big|_1^{\sqrt{2}} = -\frac{1}{4}\left[\frac{\sqrt{4-(\sqrt{2})^2}}{\sqrt{2}} - \frac{\sqrt{4-1}}{1}\right]$$

$$= -\frac{1}{4}\left[\frac{\sqrt{4-2}}{\sqrt{2}} - \sqrt{3}\right]$$

$$= -\frac{1}{4}(1-\sqrt{3})$$

$$= (\sqrt{3}-1) \quad \text{※}$$

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## EXTRA PROBLEMS

Book – H. Anton

page : 513

PROBLEMS: 1 – 25

* Evaluate $\int_0^3 f(x)\,dx$ where $f(x) = \begin{cases} x^2, & x < 2 \\ 3x-2, & x \geq 2 \end{cases}$

Sol$^n$

$$\int_0^3 f(x)\,dx = \int_0^2 f(x)\,dx + \int_2^3 f(x)\,dx$$

$$= \int_0^2 x^2\,dx + \int_2^3 (3x-2)\,dx$$